# Software Requirements Specification (SRS)

## For: Online Quiz System

### 1. Introduction

#### 1.1 Purpose

The purpose of this document is to define the software requirements for the Online Quiz System. It specifies the system's functional and non-functional requirements, architectural constraints, and security objectives.

#### 1.2 Document Conventions

This document follows the standard IEEE format adapted for Markdown. Requirements are uniquely identified (e.g., FR1, NFR1) for clear traceability.

#### 1.3 Intended Audience

This document is intended for software developers, QA engineers, project managers, and academic stakeholders involved in the system's design and deployment.

#### 1.4 Project Scope

The Online Quiz System is a secure, web-based platform that allows teachers to create and manage academic quizzes, and enables students to take quizzes and view automated grading results instantly.

## 2. Overall Description

The system operates using three primary actor roles: Students, Teachers, and Administrators. It provides a scalable platform to facilitate remote and in-person digital assessments.

## 3. Specific Requirements

### 3.1 External Interface Requirements

- **User Interfaces:** The system shall provide a responsive web interface accessible via standard web browsers (Chrome, Firefox, Edge, Safari).
- **Software Interfaces:** The frontend application shall communicate with a backend relational database via standard RESTful JSON APIs.

### 3.2 Functional Requirements (FR)

- **FR1 - User Authentication:** The system shall allow users to register and securely log in using an email and password.
- **FR2 - Quiz Creation:** A user with the 'Teacher' role shall be able to create a quiz by entering a title, instructions, and multiple-choice questions.
- **FR3 - Quiz Taking:** A user with the 'Student' role shall be able to select an active quiz, answer questions, and submit the quiz within a designated time limit.
- **FR4 - Automated Grading:** The system shall automatically calculate the student's final score immediately upon quiz submission by comparing student answers to the correct answer key.

### 3.3 Non-Functional Requirements (NFR)

- **NFR1 - Performance:** The system shall process quiz submissions and return a graded result in less than 2.0 seconds under normal server load.
- **NFR2 - Scalability:** The system shall support up to 500 concurrent students taking and submitting quizzes simultaneously without performance degradation.

- **NFR3 - Usability:** The user interface shall be intuitive, requiring less than 15 minutes of training for a new Teacher to learn how to create a quiz.

### 3.4 Security Section

**Objectives:**

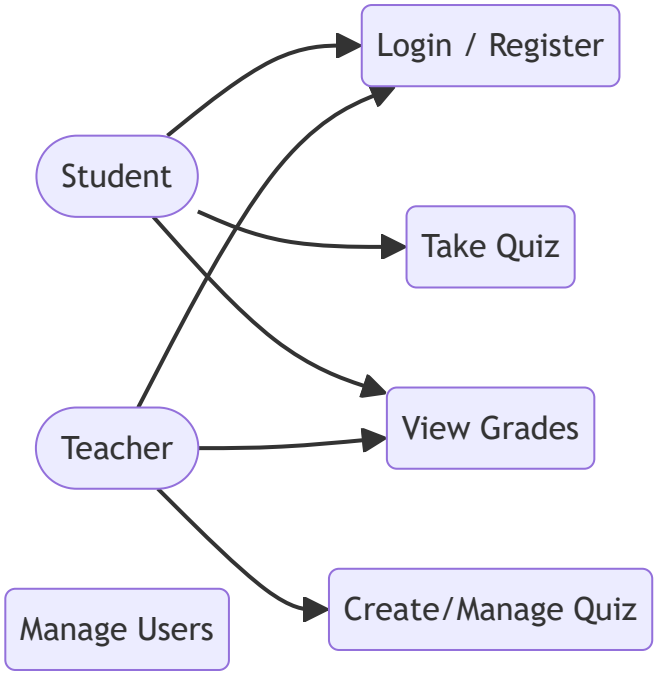
1. Maintain the complete confidentiality of student scores and private user identifying data.
2. Ensure the strict integrity of quiz questions, answer keys, and test submissions to prevent cheating or unauthorized modification.

**Requirements:**

- **SEC1 - Password Hashing:** The system shall store all user passwords in the database using a strong, salted cryptographic hash (e.g., bcrypt), ensuring passwords are never stored or transmitted in plain text.
- **SEC2 - Role-Based Access Control (RBAC):** The system shall enforce strict access controls on the backend APIs, actively preventing users with 'Student' roles from accessing endpoints designated for 'Teacher' or 'Admin' operations.
- **SEC3 - Encryption in Transit:** All data transmitted between the client interface and backend servers shall be encrypted using TLS 1.2 or higher (HTTPS).

## 4. System Models

### 4.1 UML Use Case Diagram


Login / Register
Student
Take Quiz
View Grades
Teacher
Create/Manage Quiz
Manage Users